TokenCut: Extensions for Unsupervised Object Discovery

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TokenCut: Step 1 — Graph Construction

- A fully connected undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is built over image patches.
- Nodes $\mathcal{V}$: patch feature vectors from DINO ViT-S/16 (last layer).
- Edge labels represent a similarity score; edges are thresholded at τ:

$$\mathcal{E}_{i,j} = \begin{cases} 1, & \text{if } S(\mathbf{v}_i, \mathbf{v}_j) \geq \tau \\ \epsilon, & \text{else} \end{cases} \qquad S(\mathbf{v}_i, \mathbf{v}_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\|_2 \|\mathbf{v}_j\|_2}$$

- Spatial location is implicitly encoded via positional encoding in the transformer.

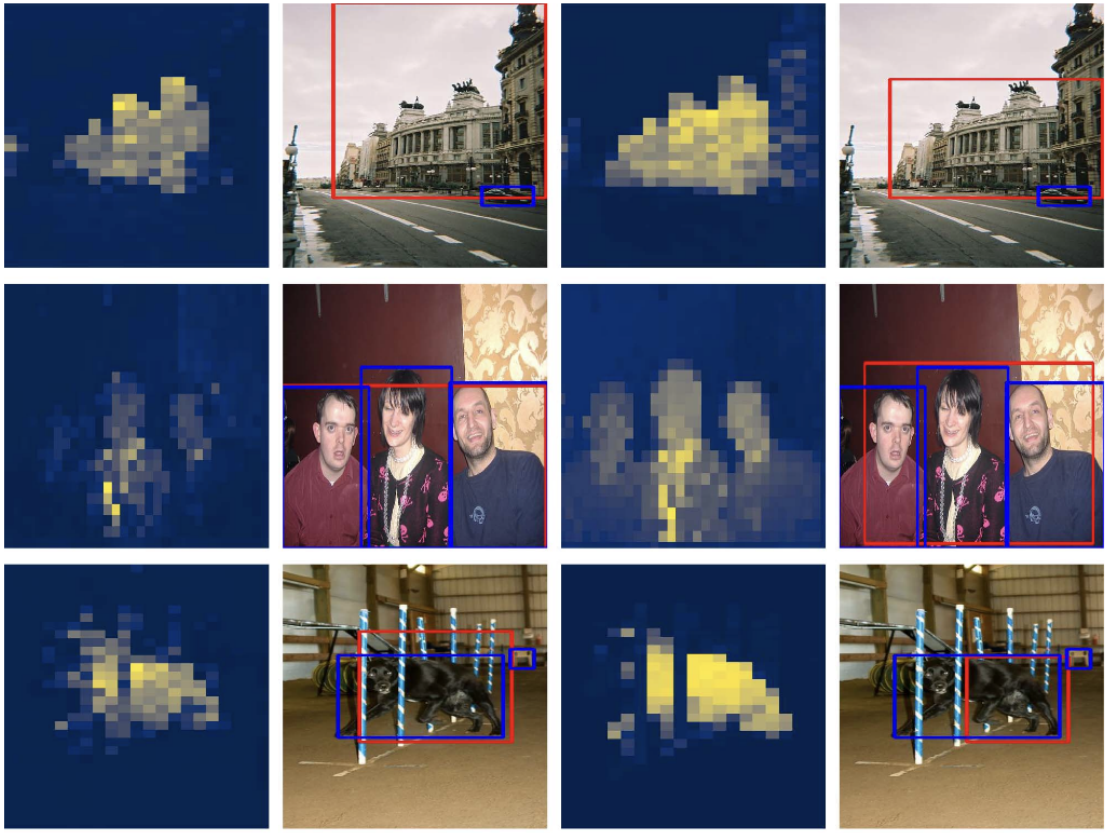
TokenCut: Step 2 — Graph Cut

- NCut computes the second smallest eigenvector of the generalised eigensystem:

$$(\mathbf{D} - \mathbf{E})\,\mathbf{y} = \lambda\,\mathbf{D}\,\mathbf{y}$$

- This eigenvector serves as *eigen-attention* highlighting salient object regions.
- **Bi-partition:** split at mean $\bar{y}_1$: $\mathcal{A} = \{\mathbf{v}_i \mid y_i \leq \bar{y}_1\}$, $\mathcal{B} = \{\mathbf{v}_i \mid y_i > \bar{y}_1\}$
- Foreground = partition with maximum absolute eigenvector value $\mathbf{v}_{\max}$ (foreground patches are less connected to the full graph: $d_i < d_j$).

Limitations of TokenCut



(a) LOST Inverse Attn. (b) LOST Detection (c) Our Eigen Attention (d) Our Detection

Fig. 8. *Failure cases on VOC12 (1st and 2nd row) and COCO (3rd row).* LOST [16] mainly relies on the map of inverse degrees (a) to perform detection (b). For our approach, we illustrate the eigenvector in (c) and our detection in (d). Blue and Red bounding boxes indicate the ground-truth and the predicted bounding boxes respectively.

Motivation

- TokenCut performs well in clean, single-object scenarios but the above failure modes limit its applicability in real-world images.
- **Goal:** Targeted extensions that address these limitations while preserving the core NCut pipeline.
- **Two directions explored:**
  1. **Occlusion handling** — Graph Diffusion, Spatial Affinity Term
  2. **Multi-object detection** — Recursive detection and combined variants
- **Code:** Core TokenCut pipeline from the official GitHub repository; all extensions and modifications coded from scratch.

Method 1: Graph Diffusion

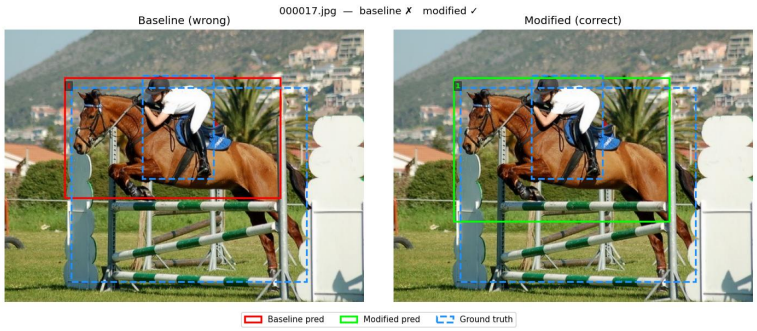
- **Idea:** Two-hop connections in the graph bridge object parts separated by occluder patches.
- **Modified affinity:**

$$E' = E + \gamma E^2, \quad E^2_{ij} = \sum_k E_{ik} E_{kj}$$

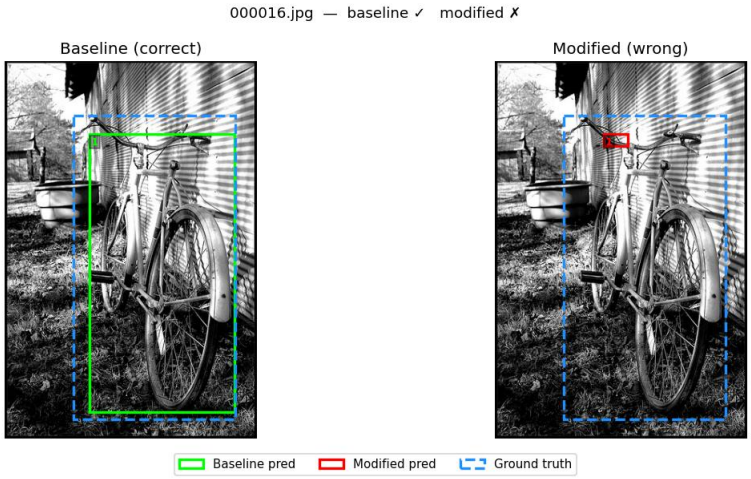
- Indirect similarity through intermediate patch k compensates for broken direct edges.
- Hyper-parameter γ controls the diffusion strength.

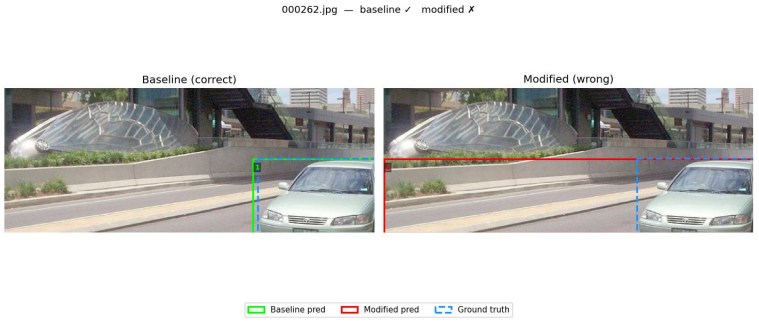
Method 1: Graph Diffusion — Results

Extension performs better



Base performs better





Method 2: Spatial Affinity Term

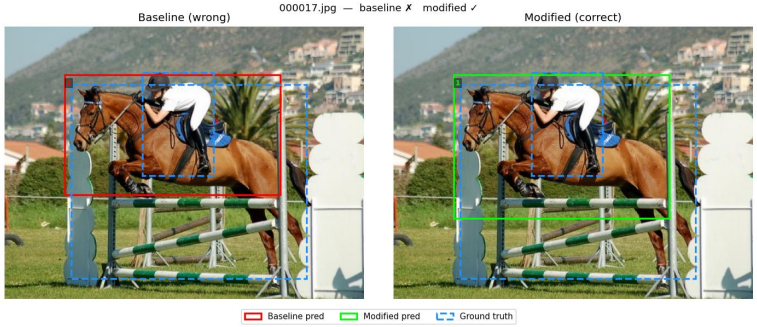
- **Idea:** Spatially adjacent patches should stay connected regardless of appearance changes.
- **Spatial term:**

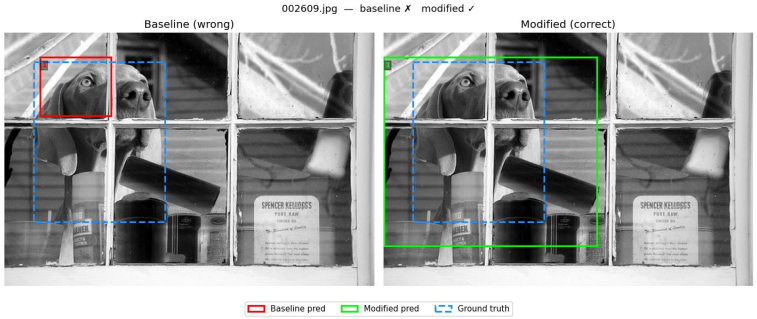
$$S_{ij} = \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma^2}\right)$$

- **Combined affinity:** $A_{ij} = (1 - \alpha)\, W_{ij} + \alpha\, S_{ij}$
- Spatial smoothness prevents over-fragmentation of the affinity graph under local occlusion.

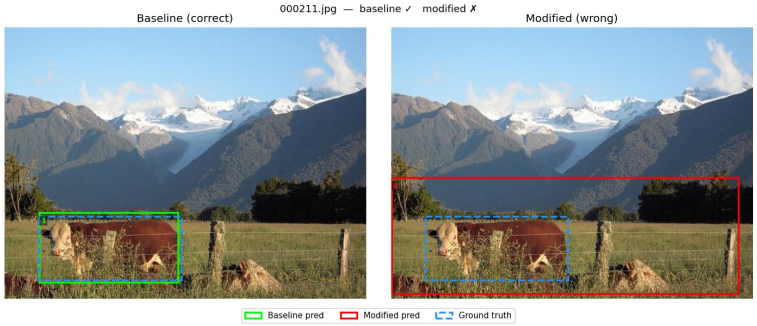
Method 2: Spatial Affinity Term — Results

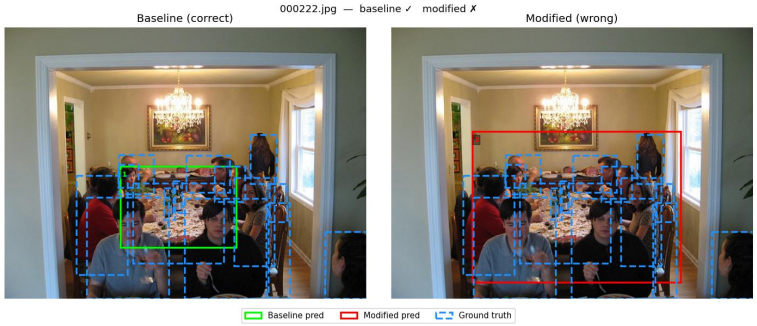
Extension performs better





Base performs better

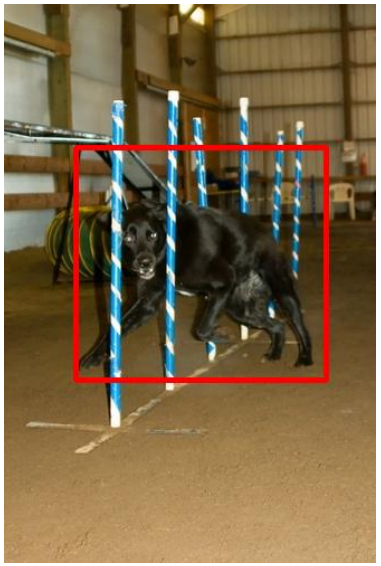




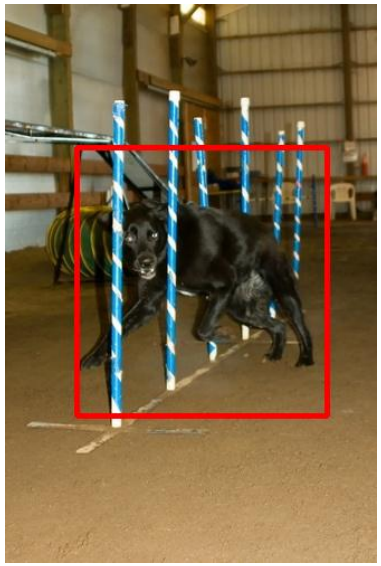
Qualitative Comparison — Occlusion Methods

Original

Baseline



Graph Diffusion



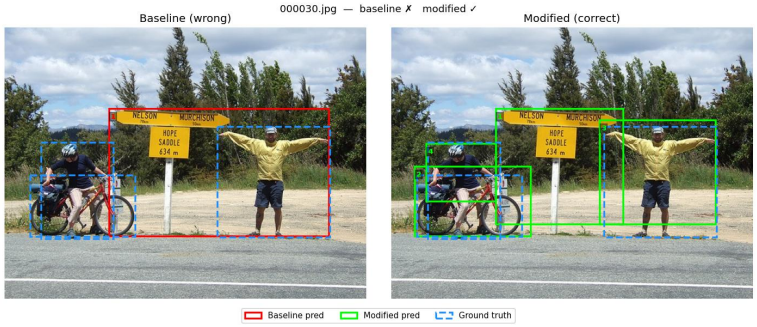
Spatial Affinity

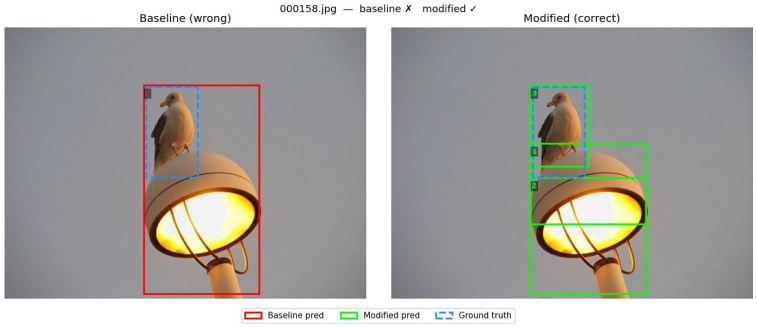
Method 3: Recursive Detection

- **Idea:** Detect one object per iteration; mask it out and re-run TokenCut on the residual graph.
- **Algorithm:**
  1. Run TokenCut $\rightarrow$ foreground mask M_k.
  2. Remove M_k patches from graph; re-run NCut on remaining nodes.
  3. Terminate when $\lambda_2 > \lambda_{\text{stop}}$ (no clear partition remains).
- Sequentially discovers multiple objects without altering the core bipartition.

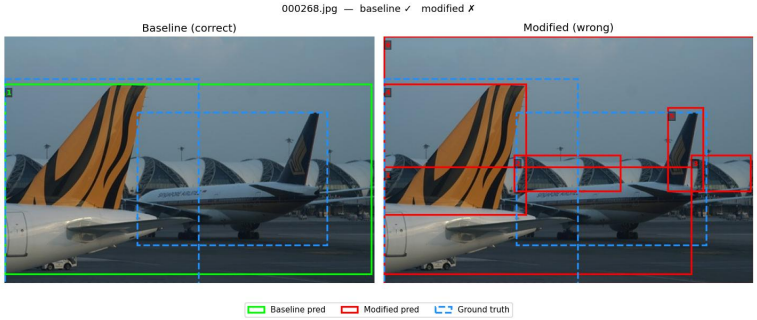
Method 3: Recursive Detection — Results

Extension performs better



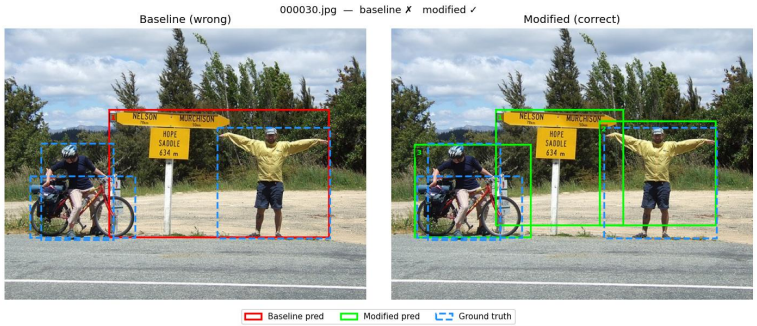


Base performs better



Method 4: Recursive + Spatial Affinity — Results

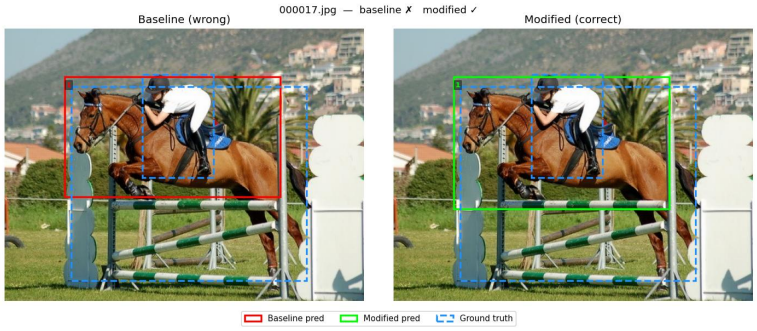
Extension performs better

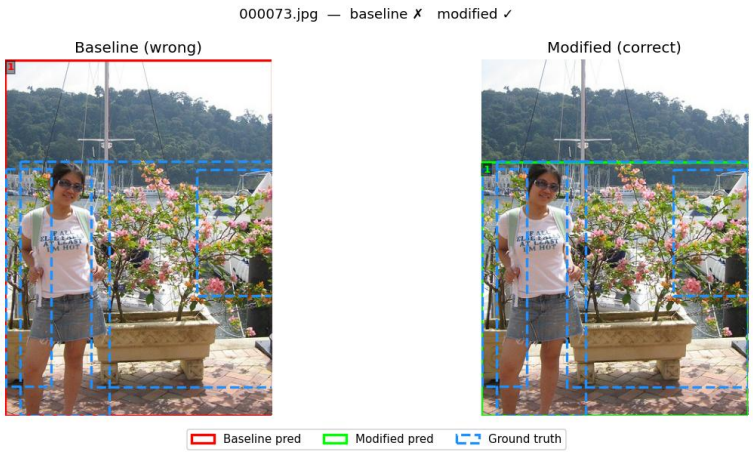


Base performs better

Method 5: Recursive + Diffusion — Results

Extension performs better





Base performs better

Quantitative Results

Dataset	Baseline	Diffusion	Spatial	Recursive	Rec.+Spatial	Rec.+Diffusion
VOC07	68.71	66.71	66.79	**72.90**	70.50	68.77
VOC12	72.07	69.27	71.12	**75.51**	75.23	71.41

Metric: CorLoc (%). Unsupervised single object discovery on VOC07 and VOC12. **Bold** = best.

Thank You!